

EXP NO.8

PRODUCER CONSUMER USING SEMAPHORES

```
sanjayganesh66@localhost:~  
[sanjayganesh66@localhost ~]$ vi procon.c  
[sanjayganesh66@localhost ~]$ gcc procon.c -o procon  
[sanjayganesh66@localhost ~]$ ./procon  
Produced: 0  
Consumed: 0  
Produced: 1  
Consumed: 1  
Produced: 2  
Produced: 3  
Consumed: 2  
Produced: 4  
Produced: 5  
Consumed: 3  
Produced: 6  
Produced: 7  
Consumed: 4  
Produced: 8  
Produced: 9  
Consumed: 5  
Consumed: 6  
Consumed: 7  
Consumed: 8  
Consumed: 9  
[sanjayganesh66@localhost ~]$
```

```
sanjayganesh66@localhost:~ — vi procon.c  
#include <pthread.h>  
#include <semaphore.h>  
#include <unistd.h>  
#define BUFFER_SIZE 5  
int buffer[BUFFER_SIZE];  
int in = 0, out = 0;  
sem_t mutex, empty, full;  
void* producer(void* arg) {  
    int item;  
    for(int i = 0; i < 10; i++) {  
        item = i;  
        sem_wait(&empty);  
        sem_wait(&mutex);  
        buffer[in] = item;  
        printf("Produced: %d\n", item);  
        in = (in + 1) % BUFFER_SIZE;  
        sem_post(&mutex);  
        sem_post(&full);  
        sleep(1);  
    }  
}  
  
void* consumer(void* arg) {  
    int item;  
    for(int i = 0; i < 10; i++) {  
-- INSERT --
```

```
sanjayganesh66@localhost:~ — vi procon.c
void* consumer(void* arg) {
    int item;
    for(int i = 0; i < 10; i++) {
        sem_wait(&full);
        sem_wait(&mutex);
        item = buffer[out];
        printf("Consumed: %d\n", item);
        out = (out + 1) % BUFFER_SIZE;
        sem_post(&mutex);
        sem_post(&empty);
        sleep(2);
    }
}

int main() {
    pthread_t p, c;

    sem_init(&mutex, 0, 1);
    sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);

    pthread_create(&p, NULL, producer, NULL);
    pthread_create(&c, NULL, consumer, NULL);

    pthread_join(p, NULL);
    pthread_join(c, NULL);
-- INSERT --
```